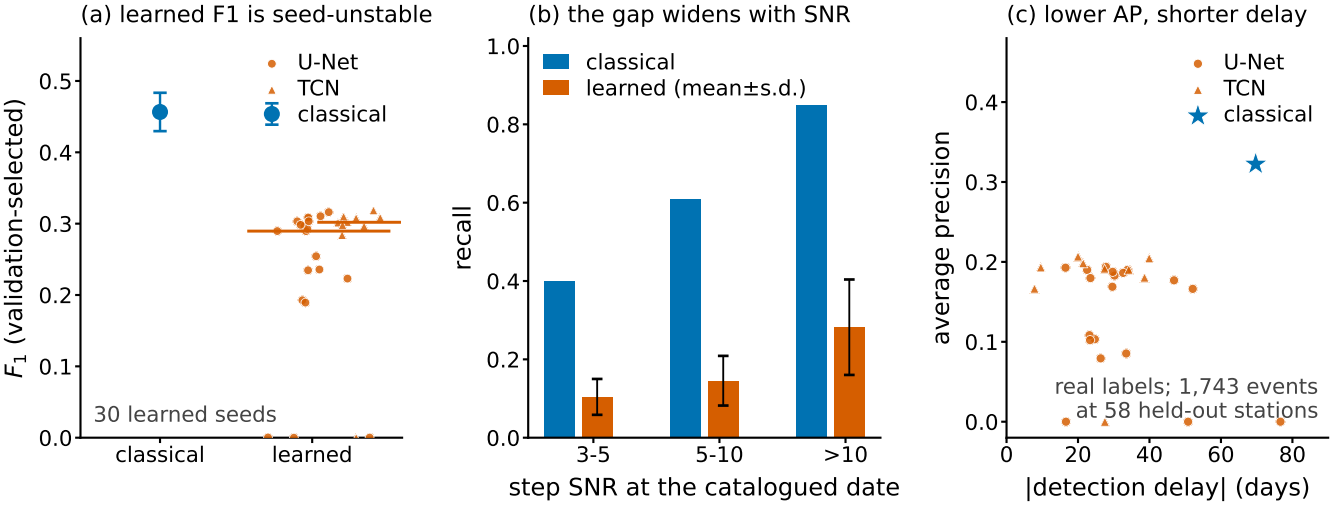
(a) learned F1 is seed-unstable
U-Net
TCN
classical
F_1 (validation-selected)
30 learned seeds
classical
learned
(b) the gap widens with SNR
classical
learned (mean±s.d.)
recall
3-5
5-10
>10
step SNR at the catalogued date
(c) lower AP, shorter delay
U-Net
TCN
classical
average precision
real labels; 1,743 events
at 58 held-out stations
|detection delay| (days)